Lesson 6 – Calculus, Derivatives, and Gradients

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Derivatives

The derivative of y with respect to x is:

$$\frac{dy}{dx}$$

The partial derivative of f with respect to x is:

$$\frac{\partial f}{\partial x}$$

Gradients

The gradient of the loss function $\mathcal{L}$ with respect to parameter θ is:

$$\nabla_\theta \mathcal{L}$$

The gradient descent update rule is:

$$\theta = \theta - \alpha \nabla_\theta \mathcal{L}$$

where α is the learning rate.

Limits

The limit of $f(x)$ as x approaches a is:

$$\lim_{x \to a} f(x)$$

As x approaches infinity:

$$\lim_{x \to \infty} f(x)$$

As x approaches zero:

$$\lim_{x \to 0} f(x)$$